

MDP Training Workshop: Part 3: User-defined Grids, User-defined Reports

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Stephen Griffith, mesonic International

MDP Workshop Agenda

- ✓ Create new WinLine report forms
- ✓ Print out data from a WinLine window, e.g., a grid control, to the new WinLine report form with the CWLReport objects
- ✓ Obtain SQL connection information with CWLDBConnection object methods
- ✓ Perform SQL Select statement with CWLDBConnection object methods
- ✓ Open WinLine database tables with CWLTable methods
- ✓ Read, write and delete data to a WinLine database table with the CWLTable object methods

Reports

- ✓ WinLine reports can be output with MDP scripts.
- ✓ Both existing forms as well as new user-defined forms can be used for output
- ✓ New forms are created in CWLPDFE
- ✓ New user-defined forms apply PDI "P99WUSERDEFINED"
- ✓ Reports are generated from the CwlWindow Class and closed from the class
 - ✓ CwlReport **CreateReport** (short Type, BSTR Name, VARIANT left, VARIANT top, VARIANT width, VARIANT height, VARIANT Description, VARIANT SpoolfileName)
 - ✓ This function creates a CwlReport object, with which a report can be output.
 - ✓ void **CloseReport** (CwlReport Report)
 - ✓ This function closes the report object. The report can no longer be used for further output.

Reports – CWLReport Class

- ✓ Properties:
 - ✓ **BSTR Name**
 - ✓ Name of report
 - ✓ **short Type**
 - ✓ Output type of report:
 - 1... To screen
 - 2... To printer
 - 4... To spooler
 - ✓ **BSTR HeaderFlags**
 - ✓ The "flags" that are currently active for the header and footer.
 - ✓ **BSTR MiddleFlags**
 - ✓ The "flags" that are currently active for the form middle section.
 - ✓ **short MultilinesLeft**
 - ✓ Number of lines of a multi-line text that will be printed on next page due to page break.

Reports – CWLReport Class

✓ BSTR **Title**

- ✓ Form title (name) (max. 50 characters) that is saved in the spool file.

✓ BSTR **Description**

- ✓ The report description (max. 100 characters) that is saved in the spool file and is shown, for example, in the grid of printed documents in the Despooler window.

✓ BOOL **ShowAbortWin**

- ✓ This property determines whether a small window with the printing progress is displayed, including an option to cancel the print out.

✓ DWORD **Id**

- ✓ This property is a number that is unique for each report while the program is running.

✓ BOOL **EnableDrilldown**

- ✓ When the property is set to TRUE, drill-down entries are active for mouse clicks.

Reports – CWLReport Class

- ✓ Methods:
 - ✓ BOOL **Header** (VARIANT Flags)
 - ✓ The header section of the report description is printed.
 - ✓ short **Middle** (VARIANT Flags)
 - ✓ The middle section of the report description is printed with this method.
 - ✓ BOOL **Footer** (VARIANT Flags)
 - ✓ The footer of the report description is printed.

Reports – CWLReport Class

- ✓ Events:
 - ✓ **OnPrintDrilldownItem** (int ReportId, ICwlEventResult *DrillDownText, short View, short Var, BSTR ItemText)
 - ✓ Event is fired when an entry that is coded as a drill-down element is clicked on in a report.
 - ✓ **OnCancel** (int ReportId, ICwlEventResult *MayClose)
 - ✓ This event is fired when the STOP button is pressed in the report or the report window is closed ("red X").
 - ✓ **OnDrilldown** (int ReportId, BSTR DrilldownText, BSTR Text)
 - ✓ This event is fired when the user clicks on a drill-down element.

Reports – CWLReport Class

- ✓ Example 1.1:
 - ✓ The contents of the entry grid in window "Short Texts" will be printed out in a WinLine report.
 - ✓ Output can be either to the screen or a printer
- ✓ Procedure:
 - ✓ A new button of type "Menu Button" is inserted in the window for a user group in CWLCTK
 - ✓ When IDDRUCKEN is selected as "Predefined Buttons" for a "Menu Button", the output options for screen or to printer are automatically made available!
 - ✓ A new report PDB is created in CWLPDFE named:
 - ✓ P99WSHORTTEXTS
 - ✓ Modify the existing CTK window script to print the contents of the window grid to the screen or printer and to set the option for closing the report output window separately when output is to the screen.

Database – CWLDbConnection Class

- ✓ This object describes a database connection.
- ✓ Previously there were three properties available for obtaining information on a database connection.
- ✓ Properties:
 - ✓ CWLDbConnectionType **Type**
 - ✓ Returns database type
 - ✓ BSTR **DatabaseName**
 - ✓ Returns database name
 - ✓ BSTR **ServerName**
 - ✓ Returns name of server
- ✓ The Class has now been expanded with the following methods:
- ✓ CWLSearchResult ***Select** (BSTR Statement)
 - ✓ A SQL statement is executed on the database connection

Database – CWLDbConnection Class

- ✓ CWLTable* **OpenTable** (BSTR strTableName, int ViewNumber, BSTR KeyColumn, int WindowId, VARIANT UseCompany)
 - ✓ This method opens the table with the specified name in the current database connection.
 - ✓ This function is used for tables that do not use the nomenclature of mesonic tables (Txxx).
- ✓ CWLTable* **OpenTable2** (short Number, short WindowId, VARIANT KeyColumn)
 - ✓ This method opens a table with number 'Number', whereby the table must conform to the mesonic nomenclature (Txxx, whereby xxx is a number with leading zeros).
- ✓ void **CloseTable** (CWLTable* pTable)
 - ✓ This method closes the opened table and discards the created variables.

Database – CWLDbConnection Class

- ✓ **BOOL ExecuteSQL** (BSTR Statement)
 - ✓ This method carries out a SQL statement.



Database – CWLDbConnection Class

- ✓ Example 1.2:
 - ✓ Obtain the company name of the current company from the Company Base Info table



Tables – CWLTable Class

- ✓ When a table has been opened with the CWLDbConnection class, an object of the CWLTable class is available to access the table.
- ✓ Properties:
 - ✓ **Name** (BOOL, read only)
 - ✓ Name of the table is held in this property.
 - ✓ **Valid** (BOOL, read only)
 - ✓ This property contains the information whether the table was successfully opened.
 - ✓ **MaxColIndex** (int, read only)
 - ✓ The index of the last defined column is saved in this property (this corresponds to the last of the variables that were created for this table).

Tables – CWLTable Class

- ✓ Methoden:
 - ✓ VARIANT **Value** (VARIANT column)
 - ✓ Use this method to access the variables in the table (the table column values).
 - ✓ void **Value** (VARIANT column, VARIANT newValue)
 - ✓ Use this method to change a column value.
 - ✓ BOOL **Get** (BSTR Key, VARIANT ExpandKey)
 - ✓ Use this method to read out a data record with a unique 'Key'.
 - ✓ BOOL **Update** ()
 - ✓ Use this method to update the data record last loaded.
 - ✓ BOOL **Insert** ()
 - ✓ Use this method to insert a new data record.
 - ✓ BOOL **Delete** (BSTR Key, VARIANT WhereStmt)
 - ✓ Use this method to delete one or more data records.

Tables – CWLTable Class

- ✓ The following function can be used only with user-defined tables:
 - ✓ BOOL **Update** ()
 - ✓ BOOL **Insert** ()
 - ✓ BOOL **Delete** (BSTR Key, VARIANT WhereStmt)

Tables – CWLTable Class

- ✓ Example 1.3: The last 20 journal lines from the ACC1 Journal
 - ✓ Display the last 20 ACC1 journal lines (records) in a grid in a new WinLine window
- ✓ Procedure:
 - ✓ Set up a new window ACC1 in CWLCTK for a user group
 - ✓ Create new CTK window script
 - ✓ Insert a grid control object
 - ✓ Create a new menu item in ACC1 to open the new window
 - ✓ Define and control the grid in the CTK window script
 - ✓ Read out the last 20 records of the ACC1 Journal, table T028, using the CWLtable class and populate window grid with them.

Thank you for your participation!
See you at the next seminar!